

Wen's Plaquette model

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1 Fixed Point Model: Start from Density Matrix

1.1 Local tensor for density matrix of the ground state

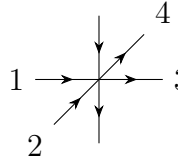
The ground state $|\psi\rangle$ for Wen's Plaquette model is stabilized by

$$X_{(i,j)}Z_{(i+1,j)}X_{(i,j-1)}Z_{(i+1,j-1)}|\psi\rangle = |\psi\rangle \quad (1)$$

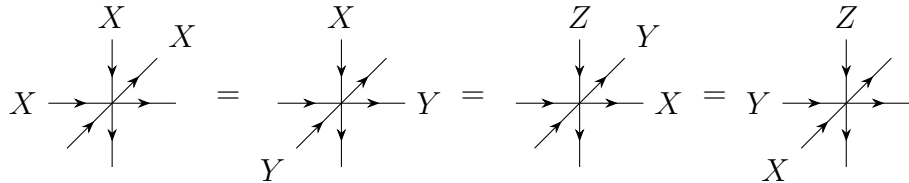
We denote the local tensor for $\rho = |\psi\rangle\langle\psi|$ to be D :

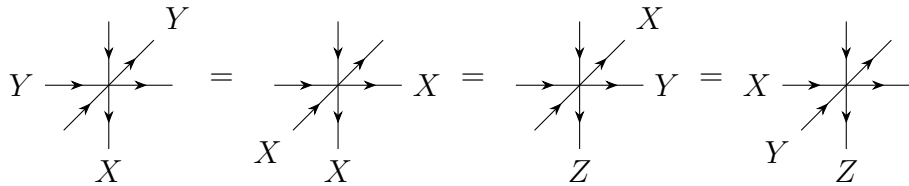
$$D = D_{\alpha\beta\gamma\delta}^{ij} |i\rangle_u \langle j|_d |\alpha\rangle_1 |\beta\rangle_2 (\gamma|_3 (\delta|_4 \quad (2)$$

where the internal legs are bosonic. The internal legs are labelled as

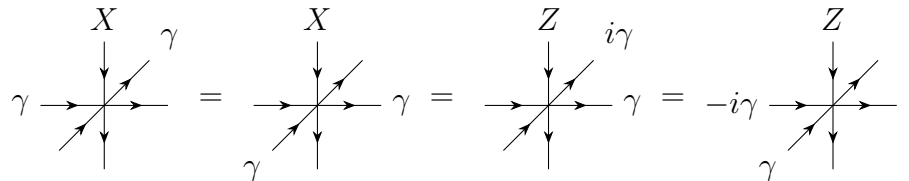

(3)

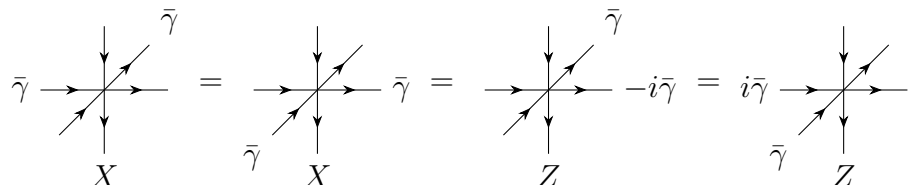
The tensor constraints are


(4)


(5)

which can be seen as the Jordan-Wigner transformed version of the following tensor constraints:


(6)


(7)

where $\gamma_i = Z_1 \cdots Z_{i-1} X_i$, $\bar{\gamma}_i = Z_1 \cdots Z_{i-1} Y_i$, $F_i = i\bar{\gamma}_i \gamma_i = Z_i$. Note that the fermionic tensor diagrams should be understood as

$$\begin{array}{c}
 X \\
 \downarrow \\
 \begin{array}{c} \nearrow \\ \rightarrow \\ \searrow \end{array} \\
 \gamma
 \end{array}
 = \gamma_2 \cdot (X_u \circ D) \cdot \gamma_3
 \quad (8)$$

rather than γ directly acting on the internal legs.

Only 6 of the equations above are independent. From these equations we can obtain 2 IGGs:

$$\begin{array}{c}
 Y \\
 \downarrow \\
 \begin{array}{c} \nearrow \\ \rightarrow \\ \searrow \end{array} \\
 X
 \end{array}
 =
 \begin{array}{c}
 X \\
 \downarrow \\
 \begin{array}{c} \nearrow \\ \rightarrow \\ \searrow \end{array} \\
 Y
 \end{array}
 \quad (9)$$

or in the fermionic version

$$\begin{array}{c}
 \gamma \\
 \downarrow \\
 \begin{array}{c} \nearrow \\ \rightarrow \\ \searrow \end{array} \\
 \gamma
 \end{array}
 =
 \begin{array}{c}
 \bar{\gamma} \\
 \downarrow \\
 \begin{array}{c} \nearrow \\ \rightarrow \\ \searrow \end{array} \\
 \bar{\gamma}
 \end{array}
 \quad (10)$$

The hermicity of ρ can be checked with

$$(D^\top)^* = W_1 W_2 D W_3^\dagger W_4^\dagger \quad (11)$$

where $(D^\top)^* = (D_{\alpha\beta\gamma\delta}^{ji})^* |i\rangle_u \langle j|_d |\alpha\rangle_1 |\beta\rangle_2 (\gamma|_3 (\delta|_4$ and

$$W = e^{-i\frac{\pi}{4}Z} \quad (12)$$

1.2 Transfer matrix

We first write down the tensor equations for $\tilde{D} = \text{Tr}_{\mathcal{H}_{\text{phy}}} D$

$$Z_1 \tilde{D} Z_4 = \tilde{D} Z_3 Z_4 = Y_1 X_2 \tilde{D} X_3 Y_4 = X_1 Y_2 \tilde{D} Y_3 X_4 \quad (13)$$

We then view the 2D system as a quasi-1D system by grouping a whole column of tensors together:

$$T_j = \text{Tr} \left[\cdots \tilde{D}_{(j,i-1)} \cdot \tilde{D}_{(j,i)} \cdot \tilde{D}_{(j,i+1)} \cdots \right] \quad (14)$$

Then we can derive the tensor equations for the transfer matrix T :

$$Z_i Z_{i+1} T = T Z_i Z_{i+1} = G T \bar{G} = \bar{G} T G \quad (15)$$

where

$$G = Y_1 X_2 \cdots Y_{L-1} X_L, \quad \bar{G} = X_1 Y_2 \cdots X_{L-1} Y_L \quad (16)$$

which satisfies $G \bar{G} = Z_1 \cdots Z_L$. In the fermionic version

$$G = i^{L/2} \gamma_1 \gamma_2 \cdots \gamma_L, \quad \bar{G} = (-i)^{L/2} \bar{\gamma}_1 \bar{\gamma}_2 \cdots \bar{\gamma}_L \quad (17)$$

1.3 Dominant eigenvectors of transfer matrix

From $Z_i Z_{i+1} T = T$ we have $Z_1 \cdots Z_L T = T$, hence $\bar{G} T G = \bar{G} Z_1 \cdots Z_L T G = G T G = T$.

We denote the dominant eigenvectors of T as $|R\rangle$ and $\langle L|$. Then we have

$$Z_i Z_{i+1} |R\rangle = |R\rangle, G |R\rangle = \pm |R\rangle \quad (18)$$

$$\langle L| Z_i Z_{i+1} = \langle L|, \langle L| G = \pm \langle L| \quad (19)$$

Then we can easily write down $|R\rangle$

$$|R\rangle = |\bar{0}\rangle \pm i^{L/2} |\bar{1}\rangle \quad (20)$$

where $|\bar{0}\rangle = |0 \cdots 0\rangle$, $|\bar{1}\rangle = |1 \cdots 1\rangle$.

Similarly we can compute $\langle L|$

$$\langle L| = \langle \bar{0}| \pm (-i)^{L/2} \langle \bar{1}| \quad (21)$$

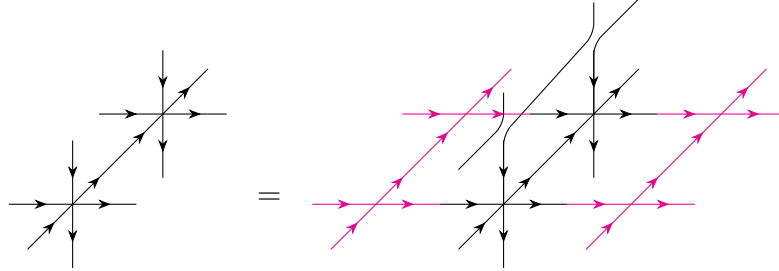
We see that they are basically cat states.

1.4 Translation symmetry

We then derive the effects of translation symmetry. We denote the symmetry action on physical legs as $\mathbf{T}_{\hat{y}}$. Since $\mathbf{T}_{\hat{y}}$ is a symmetry, we should be able to pull it down to the internal leg, which we denote the corresponding operator as $\mathbf{U}$. We should have the following

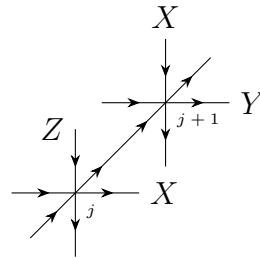
$$T = \mathbf{U} \cdot (\mathbf{T}_{\hat{y}} \circ T) \cdot \mathbf{U}^\dagger \quad (22)$$

which in diagrams



where we left the contraction of physical legs implicit.

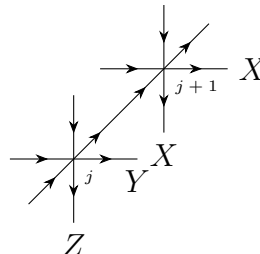
Applying the following tensor constraint



to both sides of the equation above we should have

$$X_{j-1} Y_j \cdot \mathbf{U}^\dagger = \mathbf{U}^\dagger \cdot X_j Y_{j+1} \quad (25)$$

Similarly from



we should have

$$Y_j X_{j+1} \cdot \mathbf{U}^\dagger \cdot Y_j X_{j+1} = \mathbf{U}^\dagger \quad (27)$$

It turns out that the correct $\mathbf{U}$ (modulo boundary condition) is

$$\mathbf{U} = \frac{1}{\sqrt{2}^{L-1}} \prod_{j=1}^{L-1} (1 + i X_j Y_{j+1}) \quad (28)$$

as one can directly check by calculating

$$\begin{aligned} & X_2 Y_3 \cdot \mathbf{U}^\dagger \cdot X_3 Y_4 \\ &= X_2 Y_3 \cdots (1 + i X_3 Y_4) (1 + i X_2 Y_3) \cdots X_3 Y_4 \\ &= \cdots (1 - i X_3 Y_4) (X_2 Y_3 + i) \cdots X_3 Y_4 \\ &= \cdots (X_3 Y_4 - i) (-X_2 Y_3 + i) \cdots \\ &= \cdots (1 + i X_3 Y_4) (1 + i X_2 Y_3) \cdots \\ &= \mathbf{U}^\dagger \end{aligned}$$

and

$$\begin{aligned} & Y_2 X_3 \cdot \mathbf{U}^\dagger \cdot Y_2 X_3 \\ &= Y_2 X_3 \cdots (1 + i X_3 Y_4) (1 + i X_2 Y_3) (1 + i X_1 Y_2) \cdot Y_2 X_3 \\ &= \mathbf{U}^\dagger \end{aligned}$$

Now we can calculate the disorder parameter for translation symmetry. We want to calculate $(L|\mathbf{U}|R)$. By expanding $\mathbf{U}$, we see that only terms that contribute are

$$1 \quad \text{and} \quad i^{L/2} \overline{G} \quad (29)$$

Hence we have

$$\begin{aligned} \frac{(L|\mathbf{U}|R)}{(L|R)} &= \frac{1}{2} \frac{1}{\sqrt{2}^{L-1}} \left(\langle \bar{0} | \pm (-i)^{L/2} \langle \bar{1} | \right) (1 + i^{L/2} \overline{G}) (|\bar{0}\rangle \pm i^{L/2} |\bar{1}\rangle) \\ &= \frac{1}{2} \frac{1}{\sqrt{2}^{L-1}} (2 \pm 2i^{L/2}) \\ &= \frac{1}{\sqrt{2}^{L-1}} (1 \pm i^{L/2}) \end{aligned} \quad (30)$$

which agrees with the result from Meng Cheng et al.

1.5 Disorder parameter from MPO

We repeat the calculation above using MPO.

We first write down the local tensor for $\mathbf{U} = \frac{1}{\sqrt{2}^{L-1}} \text{Tr} [U_1 \cdots U_L]$:

$$U_j = \begin{pmatrix} I & X_j \\ iY_j & Z_j \end{pmatrix} = \begin{array}{c} \text{2} \\ \downarrow \\ \text{1} \rightarrow \text{cross} \rightarrow \text{4} \\ \downarrow \\ \text{3} \end{array} \quad (31)$$

One can check that U satisfies the following tensor constraints:

$$\begin{array}{c} \text{X} \\ \downarrow \\ \text{cross} \\ \downarrow \\ \text{X} \end{array} = \begin{array}{c} \text{X} \\ \downarrow \\ \text{Z} \rightarrow \text{cross} \rightarrow \text{X} \\ \downarrow \end{array} = \begin{array}{c} \text{Y} \\ \downarrow \\ \text{Y} \rightarrow \text{cross} \rightarrow \text{Z} \\ \downarrow \\ \text{Y} \end{array} = \begin{array}{c} \text{Y} \\ \downarrow \\ \text{Y} \rightarrow \text{cross} \rightarrow \\ \downarrow \end{array} \quad (32)$$

Then we write down the local tensor for $|R\rangle$ and $\langle L|$.

$$|R\rangle = \text{Tr} [R_B R_A \cdots R_B R_A] \quad (33)$$

$$\begin{array}{c} \downarrow \\ \text{---} R_A \text{---} \\ \downarrow \end{array} = 0 \begin{array}{c} \downarrow \\ \text{---} \text{---} 0 \\ \downarrow \end{array} + i \begin{array}{c} \downarrow \\ \text{---} 1 \text{---} \text{---} 1 \\ \downarrow \end{array} \quad (34)$$

$$\begin{array}{c} \downarrow \\ \text{---} R_B \text{---} \\ \downarrow \end{array} = 0 \begin{array}{c} \downarrow \\ \text{---} \text{---} 0 \\ \downarrow \end{array} + 1 \begin{array}{c} \downarrow \\ \text{---} 1 \text{---} \text{---} 1 \\ \downarrow \end{array} \quad (35)$$

$$\langle L| = \text{Tr} [L_A L_B \cdots L_A L_B] \quad (36)$$

$$\begin{array}{c} L_A \\ \text{---} \text{---} \downarrow \end{array} = 0 \begin{array}{c} \text{---} \text{---} \downarrow \\ \downarrow \end{array} - i \begin{array}{c} \text{---} 1 \text{---} \text{---} \downarrow \\ \downarrow \end{array} \quad (37)$$

$$\begin{array}{c} L_B \\ \text{---} \text{---} \downarrow \end{array} = 0 \begin{array}{c} \text{---} \text{---} \downarrow \\ \downarrow \end{array} + 1 \begin{array}{c} \text{---} 1 \text{---} \text{---} \downarrow \\ \downarrow \end{array} \quad (38)$$

And we also have

$$Z_1 Z_2 R_A = Z_2 R_A Z_3 = Y_1 X_2 R_A X_3 \quad (39)$$

$$Z_1 Z_2 R_B = Z_2 R_B Z_3 = X_1 X_2 R_B X_3 \quad (40)$$

$$Z_1 L_A Z_2 = L_A Z_2 Z_3 = X_1 L_A X_2 Y_3 \quad (41)$$

$$Z_1 L_B Z_2 = L_B Z_2 Z_3 = X_1 L_B X_2 X_3 \quad (42)$$

Hence we can derive the tensor equation of the combined tensor BA :

$$Z_1 B A Z_4 = Z_2 B A Z_5 = Z_3 B A Z_6 = Y_2 B A Y_5 = X_1 X_3 B A X_4 X_6 = X_1 B A (i Z_4 X_4) Z_6 \quad (43)$$

$$BA = \begin{array}{c} \begin{array}{ccccc} 1 & \xrightarrow{\quad} & L_B & \xrightarrow{\quad} & L_A & \xrightarrow{\quad} & 4 \\ & & \downarrow & & \downarrow & & \\ 2 & \xrightarrow{\quad} & \boxed{U} & \xrightarrow{\quad} & \boxed{U} & \xrightarrow{\quad} & 5 \\ & & \downarrow & & \downarrow & & \\ 3 & \xrightarrow{\quad} & R_B & \xrightarrow{\quad} & R_A & \xrightarrow{\quad} & 6 \end{array} \end{array} \quad (44)$$

We can obtain

$$\begin{aligned} BA = & + |000\rangle \langle 000| \\ & + i |001\rangle \langle 001| \\ & + |010\rangle \langle 010| \\ & + i |011\rangle \langle 011| \\ & + i |100\rangle \langle 100| \\ & + |101\rangle \langle 101| \\ & + i |110\rangle \langle 110| \\ & + |111\rangle \langle 111| \end{aligned}$$

we see that

$$\mathcal{H}_{+1} = \{|000\rangle, |010\rangle, |101\rangle, |111\rangle\} \quad (45)$$

$$\mathcal{H}_{+i} = \{|001\rangle, |011\rangle, |100\rangle, |110\rangle\} \quad (46)$$

The 4-fold degeneracy can be explained by the fact that there are symmetry operators:

$$X_2, \quad X_1 X_3 \quad (47)$$

which anticommute with Z_2, Z_1

The eigenvalue 1, i may be explained by the following relation:

$$BA \cdot X_4 = iX_1 \cdot (Z_1 Z_3 BA) \quad (48)$$

Suppose $|\psi\rangle$ is the eigenket of BA with eigenvalue 1. Since $Z_1 Z_3$ commute with both $X_2, X_1 X_3$, we can assume $Z_1 Z_3 |\psi\rangle = |\psi\rangle$.

To get the exact same result with Meng Cheng et al, We need to be careful about the boundary condition of our MPO. Since our MPO comes from performing SVD to a quantum circuit, the leftmost tensor and rightmost tensor should be

$$U_1 = \begin{array}{c} \textcolor{violet}{1} \\ \downarrow \\ \textcolor{violet}{3} \leftarrow \textcolor{violet}{2} \\ \downarrow \\ \textcolor{violet}{2} \end{array} = I|0\rangle + iY|1\rangle, \quad U_L = \begin{array}{c} \textcolor{violet}{2} \\ \downarrow \\ \textcolor{violet}{1} \rightarrow \textcolor{violet}{3} \\ \downarrow \\ \textcolor{violet}{3} \end{array} = |0\rangle I + |1\rangle X \quad (49)$$

and we compute

$$\begin{array}{c} 1 \xrightarrow{\quad} \xrightarrow{L_A} \xrightarrow{L_B} \xrightarrow{\quad} 4 \\ \downarrow \quad \downarrow \\ 2 \xrightarrow{\quad} \boxed{U_L} \boxed{U_1} \xrightarrow{\quad} 5 \\ \downarrow \quad \downarrow \\ 3 \xrightarrow{\quad} \xrightarrow{R_A} \xrightarrow{R_B} \xrightarrow{\quad} 6 \end{array} = |000\rangle \langle 000| + |101\rangle \langle 101| + i|011\rangle \langle 011| + i|110\rangle \langle 110| \equiv \tilde{P} \quad (50)$$

which acts like a projector.

We also need to calculate the normalization factor $(L|R)$. So we compute

$$\begin{array}{c} 1 \xrightarrow{\quad} \xrightarrow{L_A} \xrightarrow{L_B} \xrightarrow{\quad} 3 \\ \downarrow \quad \downarrow \\ 2 \xrightarrow{\quad} \xrightarrow{R_A} \xrightarrow{R_B} \xrightarrow{\quad} 4 \end{array} = |00\rangle \langle 00| + |11\rangle \langle 11| \quad (51)$$

Hence

$$\frac{(L|\mathbf{U}|R)}{(L|R)} = \frac{1}{\sqrt{2}^{L-1}} \frac{1}{2} \text{Tr} \left[\tilde{P}(BA)^{\frac{L}{2}-1} \right] = \frac{1}{\sqrt{2}^{L-1}} \frac{1}{2} (2 + 2i^{L/2}) \quad (52)$$

Or we can still work with periodic boundary condition.

2 Beyond Fixed Point

2.1 IGG structure

Now we generalize the results above beyond fixed point model. The goal is to find the structure that produce the same disorder parameter behavior beyond fixed point.

Normally we associate one set of $\mathbb{Z}_2$ IGG with one set of toric code anyon theory: $\{1, e, m, f\}$: setting the IGG to be

$$\begin{array}{c} \text{Diagram 1: A central vertex with four legs. Top leg: arrow pointing down, labeled } J. \text{ Bottom leg: arrow pointing up, labeled } J. \text{ Left leg: arrow pointing right, labeled } J. \text{ Right leg: arrow pointing left, labeled } J. \end{array} = \begin{array}{c} \text{Diagram 2: A central vertex with four legs. Top leg: arrow pointing down. Bottom leg: arrow pointing up. Left leg: arrow pointing right. Right leg: arrow pointing left. \end{array} \quad \text{where } J^2 = 1 \quad (53)$$

With this relation, a string of J operators in the virtual space becomes topological and the endpoints are interpreted as m anyon. This is the analogue of Wilson line.

The e anyon is created by letting a tensor to be odd under the J IGG: suppose there is an operator K such that $\{J, K\} = 0$, then a tensor that is even under J IGG becomes odd when acted with K . We see that an e anyon is created by a K operator. This is the analogue of $\mathbb{Z}_2$ gauge charge.

However in Wen's Plaquette model, the analogue of K is a $\mathbb{Z}_2$ IGG itself. The IGG structure there is the following:

$$\begin{array}{c} \text{Diagram 1: A central vertex with four legs. Top leg: arrow pointing down, labeled } J_4. \text{ Bottom leg: arrow pointing up, labeled } J_2. \text{ Left leg: arrow pointing right, labeled } J_1. \text{ Right leg: arrow pointing left, labeled } J_3. \end{array} = \begin{array}{c} \text{Diagram 2: A central vertex with four legs. Top leg: arrow pointing down, labeled } K_4. \text{ Bottom leg: arrow pointing up, labeled } K_2. \text{ Left leg: arrow pointing right, labeled } K_1. \text{ Right leg: arrow pointing left, labeled } K_3. \end{array} = \begin{array}{c} \text{Diagram 3: A central vertex with four legs. Top leg: arrow pointing down. Bottom leg: arrow pointing up. Left leg: arrow pointing right. Right leg: arrow pointing left. \end{array} \quad \text{where } J^2 = K^2 = 1 \text{ and } \{J, K\} = 0 \quad (54)$$

The fact that there are 2 $\mathbb{Z}_2$ IGGs suggest that we should consider 2 copies of toric code: $\{1, e, m, f\} \times \{1, \tilde{e}, \tilde{m}, \tilde{f}\}$.

[TODO: A single J creates $m\tilde{e}$ and a single K creates $e\tilde{m}$ and the framing depends on the lattice site. We let $J_A J_B$ create mm , $J_B J_A$ create $\tilde{e}\tilde{e}$, $K_A K_B$ create $\tilde{m}\tilde{m}$, $K_B K_A$ create ee]

2.2 Translation Symmetry

Assuming that all the local tensors are the same, i.e., translation invariant and using the fact that translation of both ket and bra is equivalent to $e \leftrightarrow m, \tilde{e} \leftrightarrow \tilde{m}$, we deduce $J_A = K_B, J_B = K_A$. From now on we simply let $J \equiv J_A, K \equiv K_A$.

With the physical picture above, we can now write down how the translation of physical space acts on the entanglement space. Denote T as the transfer matrix (contracting two columns of local tensors together then trace the physical legs),

$$T = \mathbf{U} \cdot (\mathbf{T}_{\hat{y}} \circ T) \cdot \mathbf{U}^\dagger \quad (55)$$

where

$$J_i K_{i+1} \cdot \mathbf{U}^\dagger \cdot J_{i+1} K_{i+2} = \mathbf{U}^\dagger \quad (ee \text{ and } mm \text{ get translated}) \quad (56)$$

$$K_i J_{i+1} \cdot \mathbf{U}^\dagger \cdot K_i J_{i+1} = \mathbf{U}^\dagger \quad (\tilde{e}\tilde{e} \text{ and } \tilde{m}\tilde{m} \text{ stay intact}) \quad (57)$$

$$J_i J_{i+1} \cdot \mathbf{U}^\dagger \cdot (i J_{i+1} K_{i+1}) = \mathbf{U}^\dagger \quad (f_i \tilde{f}_{i+1} \rightarrow f_{i+1} \tilde{f}_{i+1}) \quad (58)$$

As we will see, (58) is the relation that contribute to the nontrivial disorder parameter. $\mathbf{U}^\dagger$ is a nontrivial $\mathbb{Z}_2$ symmetric subalgebra QCA

2.3 Hermicity (Time Reversal Symmetry)

To make sure the density matrix is hermitian, we need to impose the following:

$$T = \Theta \cdot T^* \cdot \Theta^\dagger \quad (59)$$

where

$$K^* \cdot \Theta^\dagger \cdot J = \Theta^\dagger, \quad J^* \cdot \Theta^\dagger \cdot K = \Theta^\dagger, \quad (e \leftrightarrow \tilde{e}, m \leftrightarrow \tilde{m}) \quad (60)$$

$$\Theta \cdot \Theta^* = 1 \quad (\text{ignore Kramers' degeneracy}) \quad (61)$$

2.4 Dominant Eigenvectors

The two IGGs impose two symmetry onto the dominant eigenvector:

$$J_1 K_2 \cdots J_{L-1} K_L |R\rangle = K_1 J_2 \cdots K_{L-1} J_L |R\rangle = |R\rangle$$

We make the following MPS ansatz to $|R\rangle$:

$$\begin{array}{c} \text{Diagram 1: } \text{Horizontal line with a vertical arrow pointing down labeled } R_A. \\ \text{Diagram 2: } \text{Horizontal line with a vertical arrow pointing down labeled } R_A, preceded by } \Lambda \text{ and followed by } \Lambda. \\ \text{Diagram 3: } \text{Horizontal line with a vertical arrow pointing down labeled } R_A, preceded by } V \text{ and followed by } V^*. \\ \text{Diagram 4: } \text{Horizontal line with a vertical arrow pointing down labeled } R_A, preceded by } V^* \text{ and followed by } iV\Lambda. \end{array} \quad (62)$$

$$\begin{array}{c} \text{Diagram 1: } \text{Horizontal line with a vertical arrow pointing down labeled } R_B. \\ \text{Diagram 2: } \text{Horizontal line with a vertical arrow pointing down labeled } R_B, preceded by } \Lambda \text{ and followed by } \Lambda. \\ \text{Diagram 3: } \text{Horizontal line with a vertical arrow pointing down labeled } R_B, preceded by } V^* \text{ and followed by } V. \\ \text{Diagram 4: } \text{Horizontal line with a vertical arrow pointing down labeled } R_B, preceded by } iV\Lambda \text{ and followed by } V^*. \end{array} \quad (63)$$

where

$$V^2 = \Lambda^2 = 1, \quad \{V, \Lambda\} = 0, \quad [V, V^*] = 0 \quad (64)$$

The existence of Λ is a dynamic condition to make sure that the anyons are not condensed:

$$\langle m_i m_j \rangle = \langle J_A K_B \cdots J_A K_B \rangle \rightarrow 0 \quad (65)$$

$$\langle e_i e_j \rangle = \langle J_B K_A \cdots J_B K_A \rangle \rightarrow 0 \quad (66)$$

$$\langle \tilde{m}_i \tilde{m}_j \rangle = \langle K_A J_B \cdots K_A J_B \rangle \rightarrow 0 \quad (67)$$

$$\langle \tilde{e}_i \tilde{e}_j \rangle = \langle K_B J_A \cdots K_B J_A \rangle \rightarrow 0 \quad (68)$$

This ansatz also accounts for translation invariance and time reversal symmetry:

$$e^{i\frac{\pi}{4}\Lambda} \text{Diagram 1} = \text{Diagram 2} \quad (69)$$

$$\text{Diagram 1} \text{ with } \Theta \text{ above} \text{ followed by } e^{i\frac{\pi}{4}\Lambda} = \text{Diagram 2} \quad (70)$$

2.4.1 Properties of V

From (64) we can deduce

$$V = V_1 \oplus iV_2, \quad V_i^* = V_i, \quad V_1^2 = \mathbb{I}_1, \quad V_2^2 = -\mathbb{I}_2. \quad (71)$$

The fact that the real and imaginary part of V is in different subspace comes from

$$\begin{cases} V^2 = V_1^2 - V_2^2 + i(V_1 V_2 + V_2 V_1) = \mathbb{I} \\ [V, V^*] = 2i[V_1, V_2] = 0 \end{cases} \Rightarrow V_1 V_2 = V_2 V_1 = 0$$

The fact that Λ commutes with $VV^* = \mathbb{I}_1 \oplus -\mathbb{I}_2$ tells us that Λ is also block diagonalized as $\Lambda_1 \oplus \Lambda_2$.

3 Fixed Point Model: Start from Wave Function

We can also start from the ground state wavefunction of Wen's Plaquette model:

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} = X \xrightarrow[e]{\text{Diagram 3}} X = \xrightarrow[e]{\text{Diagram 4}} Y = \xrightarrow[m]{\text{Diagram 5}} X = Y \xrightarrow[m]{\text{Diagram 6}} \quad (72)$$

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} = X \xrightarrow[m]{\text{Diagram 3}} X = \xrightarrow[m]{\text{Diagram 4}} Y = \xrightarrow[e]{\text{Diagram 5}} X = Y \xrightarrow[e]{\text{Diagram 6}} \quad (73)$$

3 of the equations above are independent. From these equations we recover the same IGG structure with (9). There are still redundancies coming from the other set of toric code. We condense one of them with the following choice

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} = Y \xrightarrow[\tilde{e}]{\text{Diagram 3}} Y \left(= \begin{array}{c} \text{Diagram 4} \\ \text{Diagram 5} \end{array} \right) \quad (74)$$

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} = X \xrightarrow[Y]{\text{Diagram 3}} X \left(= \begin{array}{c} \text{Diagram 4} \\ \text{Diagram 5} \end{array} \right) \quad (75)$$

From the equations above we can obtain the equations for reduced density matrix $\sqrt{\rho_{\partial A}}$ in entanglement space:

$$\sqrt{\rho_{\partial A}} = X_1 Y_2 \cdots X_{L-1} Y_L \cdot \sqrt{\rho_{\partial A}} = \sqrt{\rho_{\partial A}} \cdot X_1 Y_2 \cdots X_{L-1} Y_L \quad (76)$$

$$= Y_1 X_2 \cdots Y_{L-1} X_L \cdot \sqrt{\rho_{\partial A}} = \sqrt{\rho_{\partial A}} \cdot Y_1 X_2 \cdots Y_{L-1} X_L \quad (77)$$

$$= X_j Y_{j+1} \cdot \sqrt{\rho_{\partial A}} \cdot X_j Y_{j+1} \quad (78)$$

$$= Y_{2j} X_{2j+1} \cdot \sqrt{\rho_{\partial A}} = \sqrt{\rho_{\partial A}} \cdot Y_{2j} X_{2j+1} \quad (79)$$

Then we obtain the MPDO representation of $\sqrt{\rho_{\partial A}}$:

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \\ \text{Diagram 3} \\ \text{Diagram 4} \\ \text{Diagram 5} \end{array} = X \xrightarrow[\text{Diagram 6}]{\text{Diagram 7}} X = Y \xrightarrow[\text{Diagram 8}]{\text{Diagram 9}} Y = \tilde{Y} \xrightarrow[\text{Diagram 10}]{\text{Diagram 11}} Y = X \tilde{X} \xrightarrow[\text{Diagram 12}]{\text{Diagram 13}} X \quad (80)$$

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \\ \text{Diagram 3} \\ \text{Diagram 4} \\ \text{Diagram 5} \end{array} = X \xrightarrow[\text{Diagram 6}]{\text{Diagram 7}} X = Y \xrightarrow[\text{Diagram 8}]{\text{Diagram 9}} Y = Y \xrightarrow[\text{Diagram 10}]{\text{Diagram 11}} \tilde{Y} = X \xrightarrow[\text{Diagram 12}]{\text{Diagram 13}} X \tilde{X} \quad (81)$$

where the thick line represents two 2 qubits.

The MPU is again

$$\begin{array}{c} \text{---} \rightarrow \text{---} \text{---} X \\ \downarrow \\ X \end{array} = \begin{array}{c} X \\ \downarrow \\ \text{---} \rightarrow \text{---} \text{---} X \\ \downarrow \\ \text{---} \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} Z \\ \downarrow \\ Y \end{array} = \begin{array}{c} Y \\ \downarrow \\ \text{---} \rightarrow \text{---} \text{---} Y \\ \downarrow \\ \text{---} \end{array} \quad (82)$$

The tensor equations of the transfer matrix for disorder parameter are:

$$\begin{array}{c} \text{---} \rightarrow \text{---} \text{---} Y \\ \downarrow \\ Y \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} X \\ \downarrow \\ X \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} Z \\ \downarrow \\ Z \end{array} = \begin{array}{c} X \\ \downarrow \\ \text{---} \rightarrow \text{---} \text{---} X \\ \downarrow \\ \text{---} \end{array} \quad (83)$$

It is not shown in the diagrams, but the physical legs are contracted.

Explicitly, we have

$$BA = 1 + iX_1Z_2 \quad (84)$$

and the eigenvectors are:

$$\lambda_1 : \text{span}\{|+\rangle |1\rangle, |-\rangle |0\rangle\} \quad (85)$$

$$\lambda_2 : \text{span}\{|+\rangle |0\rangle, |-\rangle |1\rangle\} \quad (86)$$

$$\frac{\lambda_2}{\lambda_1} = i \quad (87)$$

Again, the source of this structure is the following relation:

$$BA \cdot \mathcal{O} = -i\mathcal{O}U_g \cdot BA, \quad \{U_g, \mathcal{O}\} = 0, \quad [BA, U_g] = 0 \quad (88)$$

Here $\mathcal{O} = X_2$, $U_g = X_1Z_2$. $U_g = -1$ on the λ_1 subspace and $U_g = 1$ on the λ_2 subspace.

Or we can also write the transfer matrix as

$$\begin{array}{c} \text{---} \rightarrow \text{---} \text{---} \tilde{Z} \\ \downarrow \\ \tilde{Y} \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} X\tilde{X} \\ \downarrow \\ X\tilde{X} \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} Z \\ \downarrow \\ X \end{array} \quad (89)$$

$$\begin{array}{c} \text{---} \rightarrow \text{---} \text{---} \tilde{Y} \\ \downarrow \\ \tilde{Y} \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} X\tilde{X} \\ \downarrow \\ X\tilde{X} \end{array} = \begin{array}{c} \text{---} \rightarrow \text{---} \text{---} Z \\ \downarrow \\ Z \end{array} \quad (90)$$

Explicitly, we have

$$AB = (1 + X_1\tilde{X}_1Z_2)(1 + iZ_1\tilde{Z}_1) \quad (91)$$

and the eigenvectors are

$$\lambda_1 : \text{span}\left\{(|+\tilde{+}\rangle - |-\tilde{-}\rangle)|0\rangle, (|+\tilde{-}\rangle - |-\tilde{+}\rangle)|1\rangle\right\} \quad (92)$$

$$\lambda_2 : \text{span}\left\{(|+\tilde{+}\rangle + |-\tilde{-}\rangle)|0\rangle, (|+\tilde{-}\rangle + |-\tilde{+}\rangle)|1\rangle\right\} \quad (93)$$

Again we have the same dominant value structure. Here $\mathcal{O} = X_1$, $U_g = Y_1\tilde{Y}_1Z_2$

4 Fixed Point Model: Start from Wave Function (fermionic)

We replace the internal legs with complex fermions in this section:

$$\begin{array}{c} \text{diagram} \end{array} = \gamma \begin{array}{c} \text{diagram} \end{array} = \begin{array}{c} \text{diagram} \end{array} \gamma = \begin{array}{c} \text{diagram} \end{array} \gamma = -i\gamma \begin{array}{c} \text{diagram} \end{array} \quad (94)$$

$$\begin{array}{c} \text{diagram} \end{array} = \gamma \begin{array}{c} \text{diagram} \end{array} = \begin{array}{c} \text{diagram} \end{array} \gamma = \begin{array}{c} \text{diagram} \end{array} \gamma = -i\gamma \begin{array}{c} \text{diagram} \end{array} \quad (95)$$

the IGGs are

$$\begin{array}{c} \text{diagram} \end{array} = \begin{array}{c} \text{diagram} \end{array} \quad (96)$$

and the redundancies are:

$$\begin{array}{c} \text{diagram} \end{array} = \bar{\gamma} \begin{array}{c} \text{diagram} \end{array} \left(= \begin{array}{c} \text{diagram} \end{array} \right) \quad (97)$$

$$\begin{array}{c} \text{diagram} \end{array} = -i\bar{\gamma} \begin{array}{c} \text{diagram} \end{array} \left(= \begin{array}{c} \text{diagram} \end{array} \right) \quad (98)$$

from which we obtain the reduced density matrix:

$$\sigma_{\partial A} = i^{L/2} \gamma_1 \cdots \gamma_L \cdot \sigma_{\partial A} = \sigma_{\partial A} \cdot i^{L/2} \gamma_1 \cdots \gamma_L \quad (99)$$

$$= i^{L/2} \bar{\gamma}_1 \cdots \bar{\gamma}_L \cdot \sigma_{\partial A} = \sigma_{\partial A} \cdot i^{L/2} \bar{\gamma}_1 \cdots \bar{\gamma}_L \quad (100)$$

$$= i\gamma_j \gamma_{j+1} \cdot \sigma_{\partial A} \cdot i\gamma_j \gamma_{j+1} \quad (101)$$

$$= i\bar{\gamma}_{2j} \bar{\gamma}_{2j+1} \cdot \sigma_{\partial A} = \sigma_{\partial A} \cdot i\bar{\gamma}_{2j} \bar{\gamma}_{2j+1} \quad (102)$$